

Ethical Challenges of Federated Learning in AI-Driven Healthcare Systems

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Abstract. This survey explores the ethical implications of Federated Learning (FL) in the healthcare sector, where the preservation of patient confidentiality is not only paramount but also legally mandated. Based on the assessment of ten key contributions to the literature, this paper maps the current FL discussion to five ethical areas: privacy and security, fairness and bias, transparency, regulatory compliance, and computational accessibility. The findings reveal that although FL presents privacy-enhanced alternatives to centralized approaches, the issues of possible information leakage, model bias, and lack of standardized regulations still remain. The paper addresses such challenges through the introduction of HEAL-FL, a theoretical multi-layer security framework that utilizes adaptive privacy measures, contribution-based trust models, and auditability through smart contracts. This research not only calls out current limitations but also lays out a systematic solution for the ethical implementation of FL in clinical AI systems.

Keywords: Federated Learning, Healthcare AI, Data Privacy, Ethical AI, Security Frameworks

1 Introduction

Federated Learning (FL) is a decentralized machine learning approach in which multiple institutions collaboratively enhance model training without the need for data exchange in raw form.[2] Unlike conventional centralized AI-based solutions, in which multiple data sources are collected and consolidated into a single repository to train models, FL keeps data on individual devices or at individual institutions and consolidates only model updates.[5] This is especially useful in medicine, where patient privacy concerns and regulatory restrictions prohibit sensitive patient information from being exchanged.

Standard centralized machine learning frameworks involve direct access to massive medical datasets in a single repository. Although this arrangement is convenient from the perspective of computational resources and operational efficiency, it creates issues related to data security, adherence to regulatory stipulations, and ethical management of patient data. FL facilitates collaborative training of AI models among hospitals, research institutions, and healthcare organizations without exposing raw patient data. This groundbreaking approach provides immense advantages in privacy and security; however, it also raises a range of ethical and technical issues, including risks of data leakage, adversarial attacks, and fairness concerns.

FL is being utilized in various healthcare applications, including diabetic retinopathy detection (e.g., Google's FL model for detecting early disease), cancer diagnosis through multi-institutional collaborations, and COVID-19 research, where FL models were employed to forecast infection trends without violating patient privacy. FL also enhances Electronic Health Records (EHR) analysis, enabling personalized treatment and risk assessment without breaching confidentiality of information.

To conduct a comprehensive comparative study of FL in healthcare in this survey, research articles are categorized into four areas as follows:

- **Privacy and Security** – Examining techniques like differential privacy, secure aggregation, and threat mitigation for preventing information leakage.
- **Fairness and Bias** – Focusing on how FL models address data heterogeneity and non-IID distributions, particularly in healthcare equity.
- **Accountability and Legal Compliance** – Assessing the degree to which models adhere to legal standards (e.g., HIPAA, GDPR) and include auditability mechanisms.
- **Accessibility and Scalability** – Considering whether FL systems support participation from low-resource institutions and offer equitable access to model performance.

This classification not only brings structure to the analysis but also surfaces the maturity and limitations of ethical thinking in Federated Learning for healthcare.

This survey aims to explore the ethical challenges of Federated Learning (FL) in medicine, specifically regarding privacy, fairness, transparency, and regulatory compliance. It seeks to identify research gaps and recommend best practices for responsible FL adoption in medical AI. More specifically, this study critically evaluates research in these areas, identifying gaps, ethical risks, and best practices. It examines how FL enhances patient privacy and reduces data leakage and adversarial attack risks, analyzes bias in FL-based AI models, and examines transparency of AI decisions in medicine. It also examines legal and ethical considerations, including

compliance with the Health Insurance Portability and Accountability Act (HIPAA), General Data Protection Regulation (GDPR), and the Personal Health Information Protection Act (PHIPA), and discusses accountability in AI-powered healthcare systems [9].

To address these ethical challenges holistically, this paper proposes a novel framework titled **HEAL-FL** (Healthcare Ethical and Adversarial-resilient Layer for Federated Learning). HEAL-FL introduces a layered architecture that integrates adaptive privacy, trust-based model updates, smart contract auditing, and open-source standardization to promote secure, fair, and regulation-compliant FL deployments in healthcare.

This paper is organized as follows: Section 2 presents an overview of Federated Learning (FL) and its applications in healthcare, along with a categorized analysis of recent research based on key ethical dimensions. Section 3 discusses the ethical challenges associated with FL in clinical contexts, including issues of privacy, fairness, accountability, and accessibility. Section 4 introduces **HEAL-FL**, a hypothetical multi-layered solution framework designed to address these challenges comprehensively. Finally, Section 5 concludes the survey by summarizing key insights and offering recommendations for ethically responsible deployment of FL in healthcare systems.

2 Related Work & Classification of FL Approaches based on Ethical Issues

The following section presents a thematically and chronologically structured summary of the most important works in FL for healthcare, classified according to the core ethical challenges - Privacy and Security, Fairness and Bias, Accountability and Legal Compliance, and Accessibility and Scalability. Instead of organizing literature by technical taxonomy, this review focuses on how each contribution addresses—or fails to address—critical ethical dimensions within FL deployments in clinical settings.

The categorization process involved close reading of the methods, discussions, and conclusions of each study. This framework provides a clear view of the evolving ethical discourse within healthcare-focused FL research and highlights the fragmented treatment of ethics across the literature. Each work was analyzed along five ethical dimensions:

Privacy and Security

Starting from the year 2020, Sheller et al. [8] explored how Federated Learning (FL) enables multi-institutional health collaborations while preserving data privacy, overcoming the limitations of centralized data-sharing (CDS). They hypothesized that FL could achieve model performance comparable to CDS approaches while maintaining institutional autonomy. Using the Brain Tumor Segmentation (BraTS) dataset [4] collected from 10 institutions, their U-Net-based convolutional neural network (CNN) model achieved 99% of the performance of its centralized counterpart. This work demonstrated how FL can improve model generalizability and reduce institutional bias, though it also highlighted concerns such as high computational requirements and residual privacy vulnerabilities. That same year, Rieke et al. [6] evaluated the broader prospects of FL in digital health, focusing on privacy preservation, data governance, and cross-institutional interoperability. Their study addressed persistent security risks including model inversion and data poisoning, and they proposed the adoption of differential privacy and secure aggregation as potential countermeasures. The paper showcased successful applications of FL in electronic health record (EHR) modeling, tumor segmentation, and diagnostic support in low-resource or rural areas. Despite these promising outcomes, they emphasized that the field still suffers from high communication overhead and the lack of standardized FL frameworks.

Building on these foundational works, Sadilek et al. [7] implemented FL across eight real-world healthcare studies, including heart failure prediction, diabetes diagnosis, and SARS-CoV-2-related risk assessment [11]. Their models demonstrated performance metrics close to centralized baselines, with AUC scores of 0.85 for heart failure (compared to 0.82) and 0.875 for diabetes (compared to 0.84). They also showed that differential privacy techniques retained 99% of performance, validating FL’s feasibility for privacy-aware AI research. However, they acknowledged that reconstruction attacks and information leakage remain a threat even with these safeguards. Later, Xu et al. [12] offered a comprehensive review of FL in healthcare, identifying privacy and communication efficiency as critical technical and ethical concerns. They detailed the performance-privacy trade-off associated with differential privacy and emphasized the scalability challenges caused by non-IID data and insufficiently standardized FL infrastructure.

Most recently, Ali et al. [1] presented a systematic literature review of FL in healthcare, outlining key challenges such as data heterogeneity, communication burden, model poisoning, and privacy breaches—particularly in EHR-based learning systems. They adhered to PRISMA guidelines to analyze research from databases like Scopus and ScienceDirect and classified FL implementations based on application domain. In terms of imaging, they highlighted how FL facilitates collaborative analysis of radiology scans such as X-rays and MRIs without transferring raw data. In disease prediction, they described FL’s use in chronic illness management (e.g., diabetes and hypertension), cancer diagnosis, and COVID-19 forecasting. The authors further advocated for robust privacy-preserving mechanisms and scalable architectures to unlock FL’s full potential in clinical practice.

Fairness and Bias

In 2023, Annapareddy et al. [3] presented a comprehensive survey focused on fairness in Federated Learning (FL) systems. They reviewed and categorized various fairness metrics (e.g., statistical parity, equal opportunity), resource allocation strategies, and fairness-aware training algorithms. Their work introduced an updated taxonomy of fairness in FL and proposed a generalized lifecycle of federated training, evaluation, and deployment stages. The authors emphasized that despite growing interest in ethical AI, fairness remains significantly underexplored in real-world healthcare FL deployments, particularly due to the lack of standardized enforcement mechanisms and practical validation in clinical settings.

Building upon this foundation, the same research group—later in 2023—extended their theoretical survey by implementing a fairness-aware FL (FAFL) system under realistic, non-IID healthcare data distributions [2]. In this follow-up study, they proposed and simulated FAFL mechanisms integrated with AES-GCM encryption to evaluate the privacy–fairness trade-off. Their experiments used the NIH Chest X-Ray dataset, comprising 25,000 samples across 13 diagnostic classes, for pneumonia classification. The results demonstrated up to 90% Top-1 accuracy, indicating that fairness and privacy can be jointly optimized. However, they also highlighted performance degradation and latency challenges in heterogeneous client environments, reinforcing the need for communication-efficient fairness-aware solutions in FL.

Accountability and Legal Compliance

Starting from the year 2021, Topaloglu et al. [9] conducted a focused inquiry into the ethical and regulatory foundations of Federated Learning in healthcare. Rather than proposing new FL models, their contribution lies in offering a conceptual and evaluative framework to assess fairness and accountability across participating institutions. Notably, they introduced the use of the Sorensen-Dice coefficient and the Gini coefficient as novel metrics for quantifying institutional contribution disparity in horizontally federated settings. This approach encourages more transparent reward structures and equitable model training dynamics. Beyond fairness, the authors explicitly addressed regulatory obligations, advocating for systematic alignment of FL systems with legal frameworks such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR). They emphasized that privacy-preserving mechanisms—like differential privacy and model encryption—should not only protect data but also enable traceability and auditability. Their work identifies key oversight mechanisms missing in many FL deployments, including informed consent tracking, data governance responsibilities, and audit trails for model access. As such, this study remains one of the few in the literature that bridges the technical and legal dimensions of FL governance, highlighting that without enforceable accountability protocols, the ethical deployment of FL remains incomplete.

Accessibility and Scalability

In 2023, Rahman et al. [5] reviewed the integration of FL, AI, and XAI in smart healthcare. Their bibliometric and architectural analysis addressed scalability challenges, resource imbalance, and the need for inclusive FL design. They emphasized the importance of explainability and equity in healthcare AI, particularly for remote monitoring and COVID-19 diagnostics.

This ethical organization of related works not only illustrates the historical progression of FL in healthcare but also highlights the fragmented attention across different ethical domains—reinforcing the need for integrated frameworks like **HEAL-FL**.

Table 1. Comparative Overview of Research Contributions in Federated Learning for Healthcare

Contributions	Datasets Used (with Stats)	Results/Observations	Ethical Issues
Annapareddy et al.[2]	NIH Chest X-Ray (25,000 images, 13 classes)	Achieves 90% Top-1 accuracy under heterogeneity; latency observed	Communication overhead; fairness in non-IID setting
Sadilek et al.[7]	8 studies incl. diabetes, COVID-19; AUC 0.85–0.875	Achieves near-centralized results; scalable with privacy	Privacy risks (reconstruction); limited robustness
Sheller et al.[8]	BraTS dataset across 10 institutions	FL achieves 99% of CDS performance; better generalization	Higher communication cost; residual privacy risk
Rieke et al.[6]	Various deployments and pilot studies	FL offers access equity, peer-to-peer robustness	Model inversion threats; governance and compliance gaps
Wang et al.[10]	NIH Chest X-Ray (112,000+ images)	Basis for many FL experiments; robust baseline	No FL evaluation; only used as dataset in other works

The final synthesis reveals not only areas of significant progress but also under-addressed gaps, reinforcing the need for an integrated ethical security architecture like **HEAL-FL**, as proposed in this paper. FL was introduced as a privacy-conscious counterpart to centralized machine learning with a particular applicability to

sensitive health settings. As suggested in the research examined, however, numerous technical as well as ethical restrictions continue to dominate current FL frameworks. Not only are they purely technical but are also deeply entangled with the ethical principles that underpin reliable AI, that is, privacy, security, fairness, transparency, trust, and accountability. Even with the utilization of tools such as differential privacy and secure aggregation, existing frameworks typically fail to achieve general ethical compliance in practical clinical environments. In the following, we classify and elaborate on these unresolved problems according to the ethical elements they infringe upon.

3 Ethical Challenges in Federated Learning for Healthcare Applications

As Federated Learning (FL) continues to gain traction in healthcare applications, its ethical implications warrant deeper examination. While FL promises privacy-preserving, collaborative AI development, several challenges persist across key ethical domains. This section categorizes and examines the unresolved ethical issues in current FL deployments, as identified through the surveyed literature.

Privacy and Security

The principal ethical issue for FL in healthcare systems is the persistent risk of information leakage, particularly during model updates. Despite the use of privacy-preserving techniques such as differential privacy and secure aggregation, research by Sadilek et al. [7] and Sheller et al. [8] indicates that FL models remain susceptible to attacks like inference attacks, model inversion, and data poisoning. These risks are particularly critical in the healthcare domain, where even minimal disclosure of sensitive patient information can result in ethical violations and legal repercussions. Therefore, although privacy is a core design motivation, it is only partially upheld in practice.

Fairness and Bias

One of the most persistent issues in FL is algorithmic unfairness caused by non-IID (non-independent and identically distributed) data distributions across institutions. This uneven data representation may lead to skewed model performance—favoring majority demographics or well-represented centers—at the expense of smaller or minority-serving institutions. Beyond performance inequities, Annareddy et al. [3] highlight that FL models often fail to employ formal fairness auditing techniques or corrective optimization methods such as constraint-based fairness or re-weighting. Furthermore, there is limited research on dynamic fairness enforcement throughout the FL lifecycle, such as during aggregation or personalization stages—suggesting that fairness is often treated as a secondary concern, rather than a core objective.

Trust and Model Robustness

Trust in FL systems is compromised by the lack of mechanisms to verify the authenticity and quality of client updates. This opens the door to adversarial behaviors like model poisoning, where compromised nodes can inject biased or malicious gradients into the aggregation process. Without a system of institutional trust evaluation, FL remains vulnerable to degraded model integrity and reduced stakeholder confidence. Rieke et al. [6] also note that most existing FL frameworks lack robust anomaly detection or trust weighting strategies, which are essential for ensuring model robustness and institutional accountability.

Accountability and Compliance

Although FL is conceptually aligned with privacy regulations such as HIPAA and GDPR, Topaloglu et al. [9] argue that few implementations include practical mechanisms for ensuring legal compliance. Features such as audit trails, real-time consent management, dynamic access control, and model usage logs are often absent. As a result, critical ethical principles—such as accountability, transparency, and informed consent—remain insufficiently operationalized in real-world FL deployments, particularly when models are used in cross-border or multi-stakeholder settings.

Computational Efficiency and Accessibility

Encryption-heavy protocols such as secure multiparty computation (SMPC) and homomorphic encryption, although essential for privacy, introduce high computational demands. This renders participation in FL prohibitively expensive for under-resourced healthcare facilities, such as rural clinics or small diagnostic labs. Sheller et al. [8] emphasize that such institutions either contribute minimal data or are excluded altogether, thereby amplifying systemic disparities. This challenge undermines not only accessibility but also equity, as the benefits of collaborative AI are skewed toward well-funded urban centers.

3.1 Domain-Specific Ethical Breakdown: Medical Imaging for Tumor Diagnosis

Tumor diagnosis using medical imaging represents one of the most prominent use-cases for Federated Learning (FL) in healthcare. In this domain, FL allows geographically distributed hospitals and research institutions to collaboratively train deep learning models (e.g., CNNs, U-Nets) on sensitive radiological data—such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT), or Positron Emission Tomography (PET)—without centralizing the data. This architecture supports compliance with privacy regulations while enabling data-rich model development for critical tasks such as segmentation, classification, and prognosis of brain or lung tumors. Despite these benefits, several ethical challenges arise in practice. Sheller et al. [8] demonstrate that although FL offers privacy protection at the data layer, it remains vulnerable at the model layer—specifically through gradient leakage, model inversion, and poisoning attacks during updates. This raises concerns about **privacy and security**, particularly since models trained on tumor imaging data can indirectly reveal patient-specific biomarkers or rare tumor patterns.

Another major concern is **fairness and accessibility**. Institutions with large, high-quality imaging datasets disproportionately influence the global model. Smaller hospitals or under-resourced diagnostic centers often contribute less frequent updates or updates with noisier data, leading to underrepresentation in the model’s learned features. As Annapareddy et al. [3] note, this imbalance can introduce systemic bias, reducing diagnostic accuracy for patients from these underrepresented sources.

Moreover, **transparency and accountability** remain weakly addressed in current tumor imaging FL implementations. Most models do not incorporate formal audit trails, consent-tracking mechanisms, or patient-level model interpretability—especially problematic in clinical oncology, where explainable decisions are crucial for treatment planning. Rieke et al. [6] highlight this gap, emphasizing the need for standardized compliance mechanisms, especially when FL is deployed at scale across national healthcare systems.

Lastly, there are **trust and robustness concerns**. Without formalized trust evaluation among participating nodes, adversarial hospitals or compromised devices could inject corrupted updates, weakening the model. Xu et al. [12] stress that dynamic trust calibration and anomaly detection are rarely integrated into FL pipelines, making current deployments fragile in adversarial settings.

In summary, while FL shows promise for privacy-preserving tumor diagnosis in radiology, the ethical landscape reveals multiple unresolved issues—from gradient-level privacy leakage and institutional dominance to weak compliance and insufficient trust calibration. Addressing these gaps requires the design of holistic FL architectures that embed ethics and security directly into their foundations—such as the HEAL-FL framework proposed in this paper.

As evidenced by both academic literature and real-world implementations, Federated Learning (FL) in healthcare continues to function within a compromised ethical framework. Existing solutions often address isolated concerns—such as privacy preservation or communication efficiency—without adequately confronting broader, systemic issues related to fairness, trust, and accountability. This underscores the pressing need for comprehensive frameworks that embed ethical principles throughout the FL pipeline, rather than treating them as peripheral considerations.

In response to these challenges, the following section introduces HEAL-FL—a novel, multi-layered architectural paradigm designed to provide holistic ethical alignment for FL in clinical contexts. HEAL-FL integrates adaptive privacy mechanisms, reputation-based trust scoring, compliance-driven audit logging, and inclusive execution strategies, offering a cohesive solution to address privacy, fairness, transparency, and regulatory obligations across healthcare deployments.

4 Hypothetical Solution for Strengthening FL Security in Healthcare

In the previous section, we illustrated that current FL applications in medicine have the tendency to resolve ethical issues like privacy, trust, or regulations individually, instead of collectively. To resolve this challenge, we propose HEAL-FL (Healthcare Ethical and Adversarial-resilient Layer for Federated Learning), a multi-layered dynamic framework developed to tackle the biggest ethical challenges faced in FL application in healthcare settings. To demonstrate how HEAL-FL can be applied in a realistic clinical context, we consider the example of ECG anomaly detection using Federated Learning. Multiple healthcare institutions, including hospitals, clinics, and specialist centers, collaboratively train a global model while keeping sensitive ECG data locally stored. HEAL-FL orchestrates this training process by embedding ethical and secure layers at every stage. This use case not only showcases the framework’s architectural adaptability but also emphasizes its practical relevance in real-world medical collaborations. By ensuring secure model updates, fairness in institutional participation, and full regulatory compliance, HEAL-FL sets a precedent for scalable ethical AI deployments in clinical environments. Moreover, the framework is designed to be modular, allowing selective integration of its layers based on the institution’s resources and regulatory context. This makes HEAL-FL suitable for progressive adoption even in healthcare systems with varying levels of digital maturity. Figure 1 illustrates this multi-institutional collaboration.

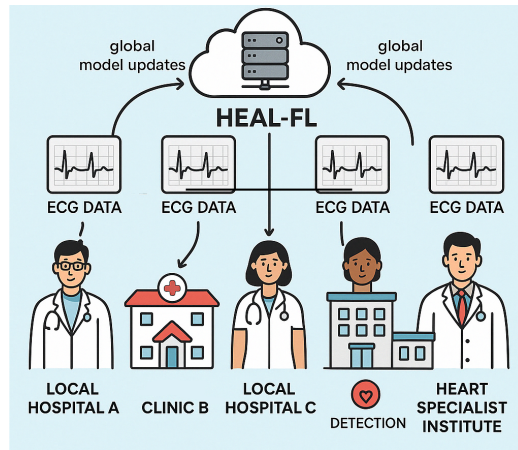


Fig. 1. Illustration of HEAL-FL in action: Local hospitals, clinics, and specialist centers contribute ECG data locally. HEAL-FL enables ethical federated learning through privacy preservation, equitable participation, and secure global model aggregation.

HEAL-FL is composed of five layers that are interconnected. Each layer plays a distinct role in the FL process so that it is safe, fair, and responsible but still functional. The system is flexible to fit into different healthcare groups—from urban hospitals to rural clinics—but still complies with the law and ethical standards. Below is an Architecture diagram of the HEAL-FL framework:

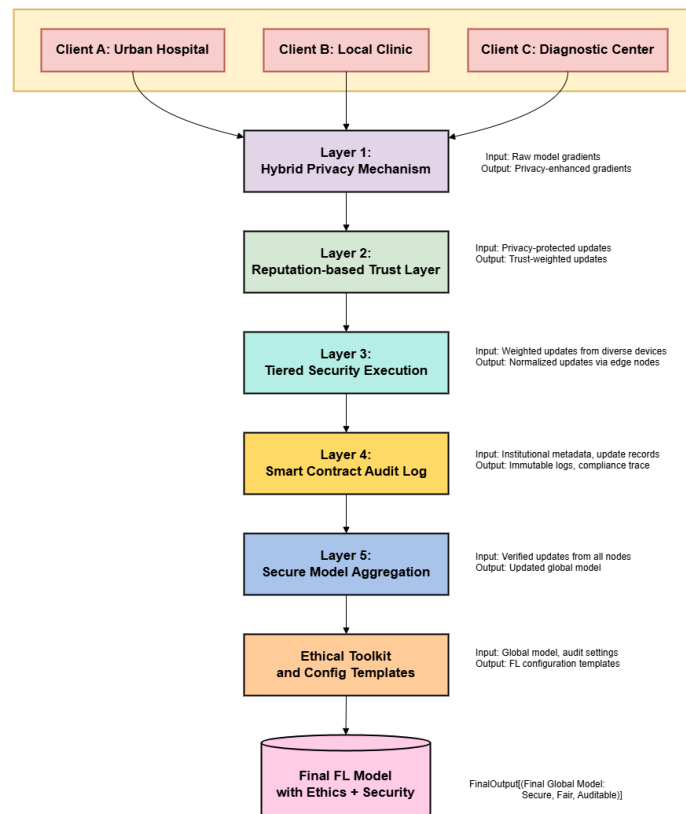


Fig. 2. Architecture Diagram of the HEAL-FL framework

Layer 1: Hybrid Privacy Mechanism

This foundational layer dynamically selects between Differential Privacy (DP) and Secure Aggregation (SA) depending on the sensitivity of the data being processed. For instance, data considered highly sensitive, such as genomic sequences or Electronic Health Records (EHRs), are protected using DP techniques that introduce calibrated noise. Meanwhile, less sensitive or already-aggregated updates may be handled through Secure Aggregation to preserve model accuracy. The primary purpose of this layer is to protect patient confidentiality while maintaining the utility of the model updates. It addresses the ethical challenge of information leakage, including risks such as model inversion attacks and privacy compromise. By integrating adaptive privacy safeguards, this layer promotes core ethical principles of privacy and security. The system receives raw model updates from local clients as input and outputs privacy-enhanced gradients that are forwarded to subsequent layers for further processing.

Layer 2: Reputation-weighted Trust Layer

This layer is designed to evaluate and weigh the trustworthiness of each participating institution by assigning dynamic trust scores. These scores are calculated using metrics such as consistency in gradient submissions, anomaly detection, and the historical reliability of contributions. The goal is to mitigate adversarial threats by down-weighting or excluding updates from clients deemed untrustworthy or malicious. Its purpose is to safeguard the model against poisoning attacks and ensure robustness against compromised clients. The ethical concerns addressed in this layer are primarily related to trust, security, and fairness, ensuring that no institution can disproportionately or maliciously influence the global model. It takes as input the identity of the client and behavioral metadata and outputs a trust-weighted score used in aggregation.

Layer 3: Tiered Security Execution System

Recognizing disparities in computational capabilities across healthcare institutions, this layer introduces a stratified execution model. Clients are categorized into security and resource tiers, with low-resource institutions (e.g., rural clinics) submitting lightweight or encrypted summaries. These updates are subsequently pre-processed at edge servers before being integrated into the model. This mechanism facilitates equitable participation without overburdening weaker clients. The main objective is to address computational overhead and ensure that system design does not unintentionally exclude under-resourced participants. The ethical values upheld here are accessibility and fairness. Input to this layer consists of simplified updates from low-resource clients, and the output is a normalized version of these updates, prepared for secure aggregation.

Layer 4: Smart Contract-based Audit Layer

Drawing inspiration from blockchain technologies, this layer ensures complete traceability of federated learning operations. It logs every operation—from model updates and privacy techniques to trust evaluations—into tamper-proof smart contracts. These records are immutable and accessible for audit by regulatory bodies, thereby fostering transparency and enabling compliance with data protection frameworks like HIPAA (Health Insurance Portability and Accountability Act), GDPR (General Data Protection Regulation), and PHIPA (Personal Health Information Protection Act). The layer’s purpose is to enforce regulatory accountability and ensure that informed consent, traceability, and institutional behavior monitoring are properly implemented. This addresses the ethical gaps surrounding auditability, legal compliance, and patient autonomy. It accepts operational metadata as input and generates immutable logs and compliance traces as output.

Layer 5: Open-source Ethical Toolkit and Standardization Layer

To facilitate widespread adoption of ethically aligned FL systems, this final layer provides an open-source toolkit with configurable templates for privacy, fairness, and auditability. It supports the implementation of predefined setups for different healthcare contexts, thus promoting rapid deployment of ethically sound systems. The purpose of this layer is to reduce the technical burden on healthcare institutions by offering customizable yet standardized configuration options. This layer addresses the challenge of absent ethical formalism and promotes governance, scalability, and fairness. The input to this layer is the globally aggregated FL model and related audit configurations, while the output consists of reusable configuration templates tailored to diverse institutional needs.

4.1 Discussion of Challenges Addressed by HEAL-FL

HEAL-FL has been conceptualized as an ethically-aware architectural framework that systematically addresses key limitations observed in current Federated Learning (FL) deployments within healthcare. It targets the issue of *privacy leakage* by employing a hybrid privacy mechanism that flexibly alternates between Differential Privacy (DP) and Secure Aggregation (SA), depending on data sensitivity. To counter threats from *malicious clients*, the framework integrates a trust-weighted aggregation strategy that evaluates and adjusts client contributions based on institutional behavior and anomaly detection.

Addressing *resource inequity*, HEAL-FL introduces a tiered security execution model that ensures equitable participation from low-resource institutions, such as rural clinics, by offloading intensive processing tasks to edge

nodes. Furthermore, the framework tackles *regulatory non-compliance* by embedding smart contract-based audit logs that document and track all federated operations, ensuring traceability and alignment with standards like HIPAA, GDPR, and PHIPA.

To resolve the *lack of transparency*, HEAL-FL incorporates immutable tracking of updates and model behaviors, which also enables effective consent management. Lastly, to mitigate the *absence of standardization*, the framework provides an open-source ethical toolkit with pre-configured templates that can be readily deployed and customized to suit various institutional and regulatory requirements. Collectively, these components establish HEAL-FL as a holistic and scalable solution to the ethical shortcomings of current FL systems in clinical settings.

Unlike many current FL deployments where ethical safeguards are retrofitted, HEAL-FL embeds these considerations directly into its core architecture. It achieves strong privacy protection, institutional trust calibration, and regulatory alignment without compromising performance. With its modular and adaptable design, the framework accommodates varying resource capabilities and legal environments across institutions. Most importantly, HEAL-FL bridges the gap between theoretical FL advancements and their ethical, practical implementation in real-world clinical settings.

While HEAL-FL proposes a comprehensive ethical and security-aware framework, its practical implementation may face certain limitations. First, the integration of blockchain-based audit layers and trust-scoring mechanisms may introduce computational overhead, especially for low-resource institutions, potentially offsetting the efficiency benefits offered by tiered execution. Second, dynamic trust models and adaptive privacy layers require extensive historical data and tuning, which may not be available in newly federated settings. Furthermore, although HEAL-FL includes standardization through open-source templates, aligning with the diverse and evolving regulatory landscapes across regions remains a challenge. Finally, the framework’s effectiveness in extreme adversarial scenarios or in the presence of coordinated malicious collusion is yet to be empirically validated, warranting future work and extensive field testing.

5 Conclusion

This paper presented an in-depth review of the ethical issues involved in FL in the healthcare industry through ten relevant academic papers and synthesized them into key ethical issues: privacy and security, fairness and bias, explainability and transparency, regulatory compliance, and computational efficiency. While FL has vast potential in enabling cooperative and privacy-preserving clinical studies, our review also points out that the area continues to be plagued with many unresolved issues—ranging from susceptibility to inference attacks and performance inconsistencies to regulatory complexities and fairness issues due to non-IID healthcare data.

Using our ethics-based classification system, we revealed that while most studies focus on privacy and security as the top priority, we found significant under-representation of studies regarding in-depth exploration of fairness or the practical challenges of applying FL in under-resourced clinical environments. Further, the absence of legacy standardized formalisms to facilitate secure and compliant FL deployments in different setups is a particularly crucial issue. Our proposed HEAL-FL approach offers a conceptual solution that combines adaptive privacy, trust weighting, distributive fairness in resource allocation, and audit trails based on smart contracts to address these issues altogether.

In summary, this survey helps to outline the current state of ethical involvement in Federated Learning medical research, while also identifying important gaps in current literature and key design imperatives that need to be addressed in future studies. In order to provide a solid foundation for medical artificial intelligence, future research needs to focus on developing robust, legally adherent, and ethically sound systems that prioritize both patient well-being and technological advancement.

While this conceptual framework outlines potential benefits through AES-GCM encryption and layered privacy safeguards, it also recognizes the need to empirically test communication latency, particularly in real-world scenarios. Future work may incorporate advanced optimization algorithms such as FedProx and broaden dataset diversity to better benchmark HEAL-FL’s practical effectiveness.

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